

Proposing an Investigation of Postoperative Hypoxemia Using the Medical Informatics Operating Room Vitals and Events Repository (MOVER)

YANG LEI, BENNY ROCHWERG, and YUNFEI YU, University of Toronto, Canada

This article proposes an investigation of postoperative hypoxemia, a post-surgical complication, using the Medical Informatics Operating Room Vitals and Events Repository (MOVER). A literature review, research questions, data mapping, and a wrangling strategy are presented.

CCS Concepts: • **Applied computing** → **Life and medical sciences**.

Additional Key Words and Phrases: Postoperative hypoxemia, MOVER, Data wrangling

ACM Reference Format:

Yang Lei, Benny Rochwerg, and Yunfei Yu. 2026. Proposing an Investigation of Postoperative Hypoxemia Using the Medical Informatics Operating Room Vitals and Events Repository (MOVER). 1, 1 (March 2026), 7 pages. <https://doi.org/10.1145/nnnnnnn.nnnnnnn>

1 Literature Review & Research Questions

Postoperative hypoxemia is a state that can arise after surgery where blood or tissue oxygen levels become deficient [6]. This state occurs regularly as a post-anesthetic care unit anesthetic complication and has a prevalence as high as 80% worldwide [6].

A previous study used logistic regressions to explore the relationship between early postoperative hypoxemia and variables such as duration of surgery, presence of heart disease, and anesthesia type [10]. Other variables, including body mass index (BMI), sex, and age, were also listed as independent variables in this study [10]. In addition, another study examined the relationship between early postoperative hypoxemia and variables including the presence of chronic disease, BMI, and smoking status using logistic regressions [12]. Furthermore, a different study employed multivariate logistic regression and found that among acute type A aortic dissection patients, BMI and gender were postoperative severe hypoxemia predictors [9]. Additional variables such as operation duration and white blood cell counts before surgery were also included in the analysis [9]. Finally, another study used logistic regression to predict postoperative hypoxemia for patients experiencing acute aortic dissection operation based on variables such as age, BMI, white blood cell count, pH, and hematocrit [8]. Other risk factors such as C-reactive protein (CRP) and lactate were also presented [8].

Although the impacts of several variables on postoperative hypoxemia were examined in these studies, they only included a maximum of four laboratory test results when modeling [8–10, 12]. This presents a research gap as including a more extensive set of laboratory test results as predictors could provide greater insight into the factors that influence the probability of postoperative hypoxemia. For instance, if postoperative hypoxemia can

Authors' Contact Information: Yang Lei, jdlyang.lei@utoronto.ca; Benny Rochwerg, benny.rochwerg@mail.utoronto.ca; Yunfei Yu, lucyyunfei.yu@mail.utoronto.ca, University of Toronto, Toronto, Ontario, Canada.

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ACM XXXX-XXXX/2026/3-ART

<https://doi.org/10.1145/nnnnnnn.nnnnnnn>

be predicted based on preoperative laboratory test results, clinicians may be able to plan accordingly to improve patient outcomes.

Consequently, we plan to answer the following research questions in our analysis:

- (1) By using patients' demographic variables (BMI, sex, and age) and latest available preoperative laboratory test results (leukocytes, pH, hematocrit, CRP, lactate, and the top five most common laboratory tests in the dataset excluding these five, namely carbon dioxide, sodium, glucose, hemoglobin, and potassium) as predictors to model the probability of postoperative hypoxemia with logistic regression, which variables are significant predictors of postoperative hypoxemia at the 0.05 level of significance?
- (2) By using only the identified significant predictors as the independent variables in another logistic regression model for the probability of postoperative hypoxemia, how accurately can the probability of postoperative hypoxemia be predicted?

2 Data Mapping

2.1 Dataset Overview

This study uses the EPIC data tables from the Medical Informatics Operating Room Vitals and Events Repository (MOVER), which contains electronic medical record data on surgeries performed at the University of California, Irvine Medical Center from 2017 to 2022 and comprises records for 64,354 surgeries and 39,685 patients [11]. This study also uses the MOVER documentation to interpret data tables and variables [1–4].

2.2 Identifiers and Unit of Analysis

The dataset includes two identifiers: MRN (medical record number, a patient-level identifier) and LOG_ID (log ID, an encounter-level identifier). A single patient (MRN) may have multiple surgical encounters, resulting in multiple LOG_ID values. Because postoperative hypoxemia and perioperative variables are encounter-specific, the analytic unit for this study is the surgical encounter, represented by LOG_ID.

2.3 Data Scope and Construct Mapping

We will use three EPIC tables from the MOVER EPIC dataset archive (EPIC_EMR.tar.gz): Patient Information (patient_information.csv), Patient Labs (patient_labs.csv), and Patient Postoperative Complications (patient_post_op_complications.csv). These tables collectively provide encounter-level demographics, preoperative laboratory test results, and the postoperative complications used to define the hypoxemia outcome.

2.4 Variable Mapping

Table 1 summarizes how each construct is represented in the available data tables. Collection Datetime (from Patient Labs) and AN_START_DATETIME (from Patient Information) are alignment timestamps used to relate laboratory measurements to the surgical timeline (i.e., identifying the most recent test results before anesthesia start). These fields will not be included as predictors in the final analytic table.

Together, these mapped fields support constructing an encounter-level dataset indexed by LOG_ID, containing the hypoxemia outcome, demographics (age, sex, and BMI), and preoperative laboratory measurements (leukocytes, pH, hematocrit, CRP, lactate, carbon dioxide, sodium, glucose, hemoglobin, and potassium).

3 Wrangling Strategy

3.1 Overview

This data wrangling process aims to prepare an analysis-ready encounter-level dataset for answering the research questions. The final dataset will contain a binary outcome variable for postoperative hypoxemia, patient

Table 1. Construct-to-Variable Mapping with Encounter Key LOG_ID and Patient Key MRN

Construct	Patient Table	Column(s)
Postoperative hypoxemia	Postoperative Complications	SMRTDTA_ELEM_VALUE
Demographics	Information	BIRTH_DATE, SEX, HEIGHT, WEIGHT
Lab test results	Labs	Lab Code, Lab Name, Observation Value, Measurement Units, Collection Datetime
Surgery (anesthesia) start	Information	AN_START_DATETIME

demographics (age, sex, and BMI), and the latest available preoperative results for 10 specific laboratory tests (leukocytes, pH, hematocrit, CRP, lactate, carbon dioxide, sodium, glucose, hemoglobin, and potassium) indexed by LOG_ID.

3.2 Source Data Tables and Relevant Features

Relevant information on patient demographics, lab test results, and postoperative hypoxemia status are obtained from three primary data tables from MOVER. The relevant variables from these tables are outlined in the Data Mapping section. All three data tables are at the encounter level and share the same unique identifier for each encounter (namely, LOG_ID).

3.2.1 Patient Postoperative Complications Table. The Patient Postoperative Complications table contains information on 64,214 unique encounters. The primary variable of interest within this table is SMRTDTA_ELEM_VALUE, which contains the specific names of postoperative complications.

3.2.2 Patient Information Table. The Patient Information table contains information on 64,354 unique encounters. The descriptions of variables of interest are listed below:

- (1) BIRTH_DATE: the age of the patient in years.
- (2) SEX: the patient's genotypical sex.
- (3) HEIGHT: the height of the patient in feet and inches (e.g., 5' 6).
- (4) WEIGHT: the weight of the patient in ounces.
- (5) AN_START_DATETIME: the start date and time of anesthesia.

We will rename the BIRTH_DATE variable to AGE for clarity.

3.2.3 Patient Labs Table. The Patient Labs table contains information for 29,079,344 lab test results across 56,125 unique encounters. The variables of interest are:

- (1) Lab Code: the laboratory test's identifier.
- (2) Lab Name: the laboratory test's descriptive common name.
- (3) Observation Value: the lab's observed value (results).
- (4) Measurement Units: the observed value's units.
- (5) Collection Datetime: the lab's collection date and time.

3.3 Data Integration and Restructuring

After selecting the relevant features, we will create three separate dataframes (one from each of the source tables):

- (1) Postoperative hypoxemia table: containing LOG_ID, MRN, and postoperative complications.
- (2) Demographics table: containing LOG_ID, MRN, age, sex, height, weight, and anesthesia start date and time.

- (3) Lab test results table: containing LOG_ID, MRN, lab code, lab name, value and units of the results, and collection date and time.

In these tables, LOG_ID uniquely identifies each encounter, and MRN uniquely identifies each patient. We identified four LOG_ID values associated with multiple MRN values, indicating single encounters linked to multiple patients. To ensure data integrity, we will exclude encounters corresponding to these four LOG_ID values so that each retained encounter corresponds to exactly one patient. As the analysis will focus on encounter-level prediction of postoperative hypoxemia, we will then merge these three dataframes into a comprehensive dataframe using LOG_ID and MRN.

3.4 Data Cleaning, Transformation, and Feature Engineering

3.4.1 Outcome Variable: Binary Indicator of Postoperative Hypoxemia. We will use the SMRTDTA_ELEM_VALUE (name of postoperative complications) variable to create the binary indicator for postoperative hypoxemia. Entries containing the keyword "hypoxemia" are classified as positive cases, while all other entries are considered negative.

The SMRTDTA_ELEM_VALUE variable has 6,741 out of 203,945 entries being non-null. We assume that these missing entries correspond to cases where no specific postoperative complications were observed. Thus, missing entries are also treated as negative cases.

To ensure the correct identification of positive cases, all text entries in this field will first be converted to lowercase. We will then create a binary variable that assigns a value of 1 to rows where the lowercase SMRTDTA_ELEM_VALUE contains the term "hypoxemia" and a value of 0 otherwise.

3.4.2 Demographic Features. We will calculate BMI as weight in kilograms divided by height in metres squared [7]. To facilitate this calculation, we will first treat the missing values of height and weight and then convert height from feet and inches into metres and weight from ounces into kilograms using the following conversion formulas:

$$\text{height in metres} = 0.0254(12(\text{feet}) + \text{inches}) \quad (1)$$

$$\text{weight in kilograms} = 0.0283(\text{weight in ounces}) \quad (2)$$

Missing values for height and weight are addressed using a two-step procedure. Because a single patient can have multiple encounters, we first attempt to fill missing heights and weights using records from the patient's nearest available encounter. If no available data exists, we plan to use regression imputation to predict the missing values based on the patient's age, sex, and whichever height or weight measurement is available.

Before attempting imputation, we will inspect the missingness patterns using visualizations such as heatmaps. If the data appears to be missing completely at random or missing not at random where the proposed imputation methods would be biased or inappropriate, we will instead retain only the encounters with no missing values for height and weight. However, if imputation is warranted, we will validate the imputation results by comparing the distributions of height and weight before and after imputation to ensure their forms and shapes remain consistent.

The SEX variable is binary with values "Female" and "Male", except for one encounter where the value is "Unknown". Since this patient has only a single recorded encounter, the missing value cannot be inferred based on the available data. We will thus drop this encounter from further analysis. We will also convert "Female" to 1 and "Male" to 0 for ease of use when modeling.

There are no missing values for the age variable.

3.4.3 Timestamps for Anesthesia and Lab Tests. The anesthesia start date and time and the lab test results collection date and time are both stored as object variables, which will need to be converted into datetime format to allow for accurate filtering and comparison during subsequent wrangling. For the same purpose, we will only include encounters with a non-missing anesthesia start date and time.

3.4.4 Lab Test Results. The Observation Value variable contains an outlier value of 9999999.0, which we will treat as missing. For Measurement Units, missing values are recorded as "Unknown" or "not reported". We will filter the merged data to exclude all entries with an Observation Value of 9999999.0, regardless of the corresponding unit's availability. We will retain results with missing Measurement Units values because certain laboratory tests, such as pH, are unitless.

We will follow the procedure below to select relevant laboratory tests.

Since each test can have multiple Lab Name values, we will choose the most common Lab Name value for each test. This will be done by filtering the merged data for entries with a Lab Name value containing the name of the desired test, ignoring case. We will then calculate the total number of entries with each distinct Lab Name value for that test and choose the Lab Name value with the largest total number of entries.

We will then choose the corresponding Lab Code value for each test by including only entries with a Lab Name value equal to the selected most common Lab Name value for that test. Then, we will calculate the total number of entries with each distinct Lab Code value corresponding to that Lab Name. We will choose the Lab Code value with the largest total number of entries. This will give us a Lab Code-Lab Name pair of values for each test (10 pairs in total). We will then filter the merged data to only include entries containing these 10 Lab Code values and these 10 Lab Name values.

For each encounter, we will restrict the merged data to only include laboratory tests with a Collection Datetime prior to the anesthesia start date and time. Finally, for each encounter, we will retain only the latest available instance for each laboratory test result prior to anesthesia.

3.5 Validating Final Dataset

We will implement basic dataset validation checks to confirm that the final dataset is analysis-ready. This includes verifying the presence of all columns required for analysis (i.e., the binary outcome, age, sex, BMI, the values and units of the 10 selected laboratory predictors, LOG_ID, and MRN) and confirming that each column has the expected data type (i.e., numeric for the binary outcome, age, sex, BMI, and the 10 laboratory value variables; character for the 10 laboratory unit variables, LOG_ID, and MRN). We will also conduct lightweight consistency checks, such as ensuring no duplicated encounter keys and screening for implausible values (e.g., negative age or BMI).

3.6 Cohort Construction and Train/Test Split

After validating the final dataset, we will construct the modeling cohort by retaining all non-duplicated encounters labeled as hypoxemia and randomly sampling an equal number of non-hypoxemia encounters to form a balanced dataset for analysis. We will then split this cohort into training (70%) and test (30%) sets. The training set will be used for exploratory data analysis (EDA) and model development, while the test set will be reserved for final model evaluation.

3.7 Exploratory Data Analysis

Using the training set only, we will generate descriptive plots to visually compare the hypoxemia and non-hypoxemia groups. These include boxplots of age, BMI, and lab predictor values by outcome group and bar charts comparing sex proportions across outcome groups. This will allow us to compare central tendency and spread between outcome groups and detect potential outliers.

If potential outliers are identified, we will assess whether they are likely data errors (e.g., unit or entry issues) or clinically plausible extreme values. Implausible values will be treated as missing and handled according to our missing data strategy, while plausible extreme values will be retained and addressed using appropriate transformations (e.g., log-transform for skewed variables) or winsorization (e.g., capping at percentile-based thresholds determined from the training set).

3.8 Conducting Analysis

For our first research question, we will fit a multivariable logistic regression model on the training set with postoperative hypoxemia as the binary outcome and the full set of candidate predictors (i.e., age, sex, BMI, and the 10 selected laboratory predictor values) as the independent variables. We will then identify the significant predictors at the 0.05 level based on the Wald test p-values for the regression coefficients.

Next, for our second research question, we will fit a reduced logistic regression model on the training set using only the significant predictors identified in our analysis for the first research question. We will then evaluate the predictive performance of the model on the test set by calculating the misclassification error rate, defined as the proportion of incorrectly classified encounters.

3.9 Writing main.py Script

We will implement a main.py script that serves as the main entry point for the project. Running main.py will call all of our other modular .py scripts in order to ingest the raw .csv files, execute the full preprocessing pipeline to generate the merged dataset, run validation checks, and export the final analytic dataset in .csv format. The script will then save EDA figures as .png files. Next, the script will save a .csv file containing the estimated coefficients and Wald test p-values from the full model for our first research question. Finally, the script will save a .txt file containing the test set misclassification error rate from the reduced model for our second research question.

Acknowledgments

The ACM Conference Proceedings Primary Article Template [5] was used to prepare this document. In addition, the authors would like to thank Professor Camellia Zakaria for providing helpful feedback.

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A GitHub Repository

This project's GitHub repository is located at <https://github.com/MOVER-analysis/MOVER-analysis>.

B Generative AI

Yang Lei used generative AI (ChatGPT) to assist with brainstorming and to refine writing. Benny Rochwerg did not use generative AI tools. Yunfei Yu used generative AI (Gemini) to refine writing.

Received 4 March 2026